A One-Step Counterexample to a Benchmark Converse for Averaging in Gradient Descent

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Abstract

We consider gradient descent with predetermined positive dimensionless stepsizes,

$$x_{k+1} = x_k - \frac{h_k}{L}\nabla f(x_k),$$

on the class of convex L-smooth functions, with an initial point within distance D of a minimizer. A natural benchmark converse says that if the last iterate does not have worst-case objective gap $LD^2/(4\sum_i h_i + 2)$, or worst-case gradient norm $LD/(\sum_i h_i + 1)$, then some nondegenerate average of the iterates should have a worst-case guarantee strictly below the corresponding benchmark. We show that this benchmark converse is false already for one gradient step. More precisely, for every one-step length $h > 3/2$, the last-iterate objective-gap worst case is strictly larger than $LD^2/(4h+2)$, but every nondegenerate average has worst-case objective gap strictly larger than $LD^2/(4h+2)$. Similarly, for every $h > \sqrt{2}$, the last-iterate gradient-norm worst case is strictly larger than $LD/(h+1)$, but every nondegenerate average has worst-case gradient norm strictly larger than $LD/(h+1)$. In particular, $h = 7/4$ gives a simultaneous counterexample while remaining below the usual $h = 2$ stability threshold.

1 Setup and benchmark converses

Fix $L > 0$, $D > 0$, and a dimension $m \geq 1$. Let $\mathcal{F}_L(\mathbb{R}^m)$ denote the class of convex differentiable functions $f : \mathbb{R}^m \to \mathbb{R}$ whose gradient is L-Lipschitz. An admissible instance is a triple

$$(f, x_0, x^\star) \quad \text{with} \quad f \in \mathcal{F}_L(\mathbb{R}^m), \quad x^\star \in \arg\min f, \quad \|x_0 - x^\star\| \leq D.$$

Denote the set of such instances by $\mathcal{I}_{L,D,m}$. In the one-step case with dimensionless stepsize $h > 0$, gradient descent produces

$$x_1 = x_0 - \frac{h}{L}\nabla f(x_0).$$

For $\sigma = (\sigma_0, \sigma_1)$ in the simplex

$$\Delta_2 = \{(\sigma_0, \sigma_1) \in \mathbb{R}^2_+ : \sigma_0 + \sigma_1 = 1\},$$

the averaged output is

$$x_\sigma = \sigma_0 x_0 + \sigma_1 x_1.$$

The last-iterate scheme is $\sigma^{\text{last}} = (0, 1)$. We call σ nondegenerate when $\sigma \neq \sigma^{\text{last}}$.

For a fixed one-step length h, define the objective-gap and gradient-norm worst cases of the reported point x_σ by

$$\mathcal{W}_{\mathrm{obj}}(\sigma;h) = \sup_{(f,x_0,x^\star)\in\mathcal{I}_{L,D,m}} \big(f(x_\sigma) - f(x^\star)\big),$$
$$\mathcal{W}_{\mathrm{grad}}(\sigma;h) = \sup_{(f,x_0,x^\star)\in\mathcal{I}_{L,D,m}} \|\nabla f(x_\sigma)\|\,.$$

We use suprema rather than maxima only to avoid unnecessary compactness issues. The counterexamples below are explicit lower-bound instances, so the same argument applies to any formulation in which these worst cases are known to be attained.

The one-step benchmark quantities are

$$B_{\mathrm{obj}}(h) = \frac{LD^2}{4h+2}, \qquad B_{\mathrm{grad}}(h) = \frac{LD}{h+1}.$$

The benchmark converses under consideration are the following implications:

$$\mathcal{W}_{\mathrm{obj}}(\sigma^{\mathrm{last}};h) \neq B_{\mathrm{obj}}(h) \quad\Longrightarrow\quad \exists\,\sigma\in\Delta_2\setminus\{\sigma^{\mathrm{last}}\} \text{ such that } \mathcal{W}_{\mathrm{obj}}(\sigma;h) < B_{\mathrm{obj}}(h),$$
$$\mathcal{W}_{\mathrm{grad}}(\sigma^{\mathrm{last}};h) \neq B_{\mathrm{grad}}(h) \quad\Longrightarrow\quad \exists\,\sigma\in\Delta_2\setminus\{\sigma^{\mathrm{last}}\} \text{ such that } \mathcal{W}_{\mathrm{grad}}(\sigma;h) < B_{\mathrm{grad}}(h).$$

These are precisely the one-step versions of the displayed benchmark converses in which $\sum_i h_i = h$.

2 Huber lower bounds for one effective step

We first record two elementary lower bounds. They use only explicit Huber functions and do not rely on any external performance-estimation theorem.

Lemma 1 (A smooth Huber family)**.** *For $L,\eta > 0$, define $\Phi_{L,\eta} : \mathbb{R}\to\mathbb{R}$ by*

$$\Phi_{L,\eta}(u) = \begin{cases} \dfrac{L}{2}u^2, & |u| \le \eta, \\ L\eta|u| - \dfrac{L}{2}\eta^2, & |u| > \eta. \end{cases}$$

Then $\Phi_{L,\eta}$ is convex and L-smooth, meaning that its derivative is globally L-Lipschitz. Consequently, for any $m \ge 1$, the function

$$f(x) = \Phi_{L,\eta}(x_1), \qquad x = (x_1,\dots,x_m)\in\mathbb{R}^m,$$

is convex and L-smooth on $\mathbb{R}^m$, and 0 is a minimizer of f.

Proof. The derivative of $\Phi_{L,\eta}$ is

$$\Phi'_{L,\eta}(u) = \begin{cases} -L\eta, & u \le -\eta, \\ Lu, & |u| \le \eta, \\ L\eta, & u \ge \eta. \end{cases}$$

Equivalently,

$$\Phi'_{L,\eta}(u) = \mathrm{proj}_{[-L\eta,L\eta]}(Lu).$$

Since projection onto an interval is nondecreasing and 1-Lipschitz, $\Phi'_{L,\eta}$ is nondecreasing and L-Lipschitz. Thus $\Phi_{L,\eta}$ is convex and L-smooth.

For $f(x) = \Phi_{L,\eta}(x_1)$, we have

$$\nabla f(x) = \Phi'_{L,\eta}(x_1)e_1,$$

where $e_1 = (1, 0, \ldots, 0)$. Hence, for all $x, y \in \mathbb{R}^m$,

$$\|\nabla f(x) - \nabla f(y)\| = |\Phi'_{L,\eta}(x_1) - \Phi'_{L,\eta}(y_1)| \leq L|x_1 - y_1| \leq L\|x - y\|.$$

Convexity of f follows from convexity of $\Phi_{L,\eta}$ and linearity of $x \mapsto x_1$. Finally, $\Phi_{L,\eta} \geq 0$ and $\Phi_{L,\eta}(0) = 0$, so 0 is a minimizer of f. In dimension $m > 1$, the minimizer need not be unique; this is immaterial. □

Lemma 2 (Huber lower bounds for one effective step)**.** *Fix $t \geq 0$. Let $e_1 = (1, 0, \ldots, 0) \in \mathbb{R}^m$, with the evident interpretation $e_1 = 1$ if $m = 1$.*

1. *There exists an instance $(f, x_0, x^\star) \in \mathcal{I}_{L,D,m}$ such that, for*

$$x^+ = x_0 - \frac{t}{L}\nabla f(x_0),$$

one has

$$f(x^+) - f(x^\star) = \frac{LD^2}{4t+2}.$$

2. *There exists an instance $(f, x_0, x^\star) \in \mathcal{I}_{L,D,m}$ such that, for the same update rule,*

$$\left\|\nabla f(x^+)\right\| = \frac{LD}{t+1}.$$

Proof. Set $x^\star = 0$ and $x_0 = De_1$. Then $\|x_0 - x^\star\| = D$.

For the objective-gap lower bound, choose

$$\eta = \frac{D}{2t+1}, \qquad f(x) = \Phi_{L,\eta}(x_1).$$

By Lemma 1, this is an admissible convex L-smooth function with 0 as a minimizer. Since $D \geq \eta$, we have $\nabla f(x_0) = L\eta e_1$. Therefore

$$x^+ = De_1 - t\eta e_1 = \frac{(t+1)D}{2t+1}e_1.$$

Moreover,

$$\frac{(t+1)D}{2t+1} \geq \frac{D}{2t+1} = \eta,$$

so x^+ lies on the affine branch of the Huber function. Hence

$$\begin{aligned}
f(x^+) - f(x^\star) &= L\eta\frac{(t+1)D}{2t+1} - \frac{L}{2}\eta^2 \\
&= L\frac{D}{2t+1}\frac{(t+1)D}{2t+1} - \frac{L}{2}\frac{D^2}{(2t+1)^2} \\
&= \frac{LD^2}{(2t+1)^2}\left(t + 1 - \frac{1}{2}\right) = \frac{LD^2}{2(2t+1)} = \frac{LD^2}{4t+2}.
\end{aligned}$$

For the gradient-norm lower bound, choose instead

$$\eta = \frac{D}{t+1}, \qquad f(x) = \Phi_{L,\eta}(x_1).$$

Again $D \geq \eta$ and $\nabla f(x_0) = L\eta e_1$. Thus

$$x^+ = De_1 - t\eta e_1 = \left(D - \frac{tD}{t+1}\right) e_1 = \frac{D}{t+1} e_1 = \eta e_1.$$

At $u = \eta$, the derivative of $\Phi_{L,\eta}$ is $L\eta$. Therefore

$$\left\|\nabla f(x^+)\right\| = L\eta = \frac{LD}{t+1}.$$

□

3 Quadratic overshoot of the last iterate

The next elementary instance shows that the last iterate can fail the benchmark identities because of quadratic overshoot.

Lemma 3 (Quadratic overshoot)**.** *Let*

$$q(x) = \frac{L}{2} x_1^2, \qquad x_0 = De_1, \qquad x^\star = 0.$$

For the one-step update

$$x_1 = x_0 - \frac{h}{L} \nabla q(x_0),$$

one has

$$x_1 = (1-h)De_1,$$

and consequently

$$q(x_1) - q(x^\star) = \frac{LD^2}{2}(1-h)^2, \qquad \|\nabla q(x_1)\| = LD|1-h|.$$

In particular,

$$\begin{aligned} h > \frac{3}{2} \quad &\Longrightarrow \quad \frac{LD^2}{2}(1-h)^2 > \frac{LD^2}{4h+2}, \\ h > \sqrt{2} \quad &\Longrightarrow \quad LD|1-h| > \frac{LD}{h+1}. \end{aligned}$$

Proof. Since $\nabla q(x) = Lx_1 e_1$, we have $\nabla q(x_0) = LDe_1$, and hence

$$x_1 = De_1 - \frac{h}{L} LDe_1 = (1-h)De_1.$$

The displayed formulas for $q(x_1) - q(x^\star)$ and $\|\nabla q(x_1)\|$ follow immediately.

If $h > 3/2$, then $h > 1$, and

$$\begin{aligned} \frac{(h-1)^2}{2} - \frac{1}{4h+2} &= \frac{(h-1)^2(2h+1) - 1}{2(2h+1)} \\ &= \frac{2h^3 - 3h^2}{2(2h+1)} = \frac{h^2(2h-3)}{2(2h+1)} > 0. \end{aligned}$$

Multiplying by LD^2 gives the objective-gap inequality. If $h > \sqrt{2}$, then $h > 1$, and

$$(h-1) - \frac{1}{h+1} = \frac{(h-1)(h+1)-1}{h+1} = \frac{h^2-2}{h+1} > 0.$$

Multiplying by LD gives the gradient-norm inequality. $\square$

4 Counterexample theorem

Theorem 1 (Failure of the benchmark converses)**.** *The benchmark objective-gap converse and benchmark gradient-norm converse are false already for $N = 1$.*

More precisely, fix a one-step length $h > 0$.

1. *If $h > 3/2$, then*
$$\mathcal{W}_{\mathrm{obj}}(\sigma^{\mathrm{last}}; h) > B_{\mathrm{obj}}(h),$$
but every nondegenerate averaging vector $\sigma \in \Delta_2 \setminus \{\sigma^{\mathrm{last}}\}$ satisfies
$$\mathcal{W}_{\mathrm{obj}}(\sigma; h) > B_{\mathrm{obj}}(h).$$
Thus the antecedent of the benchmark objective-gap converse holds, while its asserted conclusion fails.

2. *If $h > \sqrt{2}$, then*
$$\mathcal{W}_{\mathrm{grad}}(\sigma^{\mathrm{last}}; h) > B_{\mathrm{grad}}(h),$$
but every nondegenerate averaging vector $\sigma \in \Delta_2 \setminus \{\sigma^{\mathrm{last}}\}$ satisfies
$$\mathcal{W}_{\mathrm{grad}}(\sigma; h) > B_{\mathrm{grad}}(h).$$
Thus the antecedent of the benchmark gradient-norm converse holds, while its asserted conclusion fails.

Proof. Let $\sigma = (\sigma_0, \sigma_1) \in \Delta_2$. For every admissible instance,

$$x_\sigma = \sigma_0 x_0 + \sigma_1 x_1 = \sigma_0 x_0 + \sigma_1 \left(x_0 - \frac{h}{L} \nabla f(x_0) \right) = x_0 - \frac{h\sigma_1}{L} \nabla f(x_0).$$

Thus, after one gradient step, averaging is exactly the same as taking one gradient step with effective dimensionless stepsize

$$t = h\sigma_1.$$

If σ is nondegenerate, then $\sigma \neq (0,1)$, so $\sigma_1 < 1$, and hence

$$0 \leq t < h.$$

We first prove the objective claim. Assume $h > 3/2$. Applying Lemma 3 to the admissible quadratic instance gives

$$\mathcal{W}_{\mathrm{obj}}(\sigma^{\mathrm{last}}; h) \geq \frac{LD^2}{2}(h-1)^2 > \frac{LD^2}{4h+2} = B_{\mathrm{obj}}(h).$$

Thus the last-iterate objective-gap worst case is not equal to the benchmark.

Now take any nondegenerate $\sigma \in \Delta_2$. Let $t = h\sigma_1 < h$. Since x_σ is one effective step of length t, Lemma 2 gives

$$\mathcal{W}_{\text{obj}}(\sigma; h) \geq \frac{LD^2}{4t+2}.$$

The function $u \mapsto 1/(4u+2)$ is strictly decreasing on $[0, \infty)$, and $t < h$, so

$$\frac{LD^2}{4t+2} > \frac{LD^2}{4h+2} = B_{\text{obj}}(h).$$

Therefore $\mathcal{W}_{\text{obj}}(\sigma; h) > B_{\text{obj}}(h)$ for every nondegenerate σ. In particular, no nondegenerate averaging vector has objective-gap guarantee below $B_{\text{obj}}(h)$.

The gradient-norm argument is identical. Assume $h > \sqrt{2}$. The quadratic instance of Lemma 3 gives

$$\mathcal{W}_{\text{grad}}(\sigma^{\text{last}}; h) \geq LD(h-1) > \frac{LD}{h+1} = B_{\text{grad}}(h),$$

so the last-iterate gradient-norm worst case is not equal to the benchmark. For any nondegenerate σ, again set $t = h\sigma_1 < h$. Lemma 2 gives

$$\mathcal{W}_{\text{grad}}(\sigma; h) \geq \frac{LD}{t+1}.$$

Since $u \mapsto 1/(u+1)$ is strictly decreasing on $[0, \infty)$ and $t < h$,

$$\frac{LD}{t+1} > \frac{LD}{h+1} = B_{\text{grad}}(h).$$

Thus $\mathcal{W}_{\text{grad}}(\sigma; h) > B_{\text{grad}}(h)$ for every nondegenerate σ, and no nondegenerate averaging vector has gradient-norm guarantee below $B_{\text{grad}}(h)$. □

Corollary 1 (A simultaneous counterexample with $h < 2$)**.** *Take $N = 1$ and $h = 7/4$. Then both benchmark converses fail simultaneously. Explicitly,*

$$B_{\text{obj}}(7/4) = \frac{LD^2}{9}, \qquad B_{\text{grad}}(7/4) = \frac{4LD}{11}.$$

The quadratic instance has last-iterate values

$$q(x_1) - q(x^\star) = \frac{9LD^2}{32} > \frac{LD^2}{9}, \qquad \|\nabla q(x_1)\| = \frac{3LD}{4} > \frac{4LD}{11},$$

so both last-iterate benchmark identities fail. Nevertheless, every nondegenerate average $\sigma \neq (0,1)$ satisfies

$$\mathcal{W}_{\text{obj}}(\sigma; 7/4) > \frac{LD^2}{9}, \qquad \mathcal{W}_{\text{grad}}(\sigma; 7/4) > \frac{4LD}{11}.$$

Proof. The value $h = 7/4$ satisfies $h > 3/2$ and $h > \sqrt{2}$. The result follows from Theorem 1. The displayed numerical values are obtained by direct substitution into Lemma 3 and the definitions of B_{obj} and B_{grad}. □

Remark 1 (Scope of the counterexample)**.** *The theorem refutes the benchmark converse: failure of the last-iterate benchmark identity need not imply the existence of a nondegenerate average whose worst-case guarantee is strictly below the same Huber benchmark. It does not address the logically different statement that, whenever the benchmark identity fails, some average improves on the actual worst-case performance of the last iterate. In the counterexample above, the actual last-iterate worst case is larger than the Huber benchmark, and these two comparison targets should not be conflated.*

Remark 2 (Dimension)**.** *All instances used in the proof are one-dimensional, or depend only on the first coordinate. Hence the counterexample works in dimension* $m = 1$ *and embeds verbatim in every dimension* $m \geq 1$*.*

References

[1] Alan Luner and Benjamin Grimmer. *On Averaging and Extrapolation for Gradient Descent*. arXiv:2402.12493, 2025. https://arxiv.org/abs/2402.12493.